

DIALA LTEIF

AI/ML Scientist | Deep Learning | Computer Vision | Medical Imaging Applications

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SUMMARY

Machine Learning Engineer and PhD in Computer Science with expertise in Python, PyTorch, statistical analysis, computer vision, and scalable AI pipelines. Experienced in designing, training, and evaluating machine learning models on large real-world datasets, with strong skills in data processing, visualization, distributed training, and performance optimization. Proven ability to build end-to-end workflows, collaborate across interdisciplinary teams, and translate technically rigorous research into practical, high-impact solutions.

EDUCATION

PhD in Computer Science	09/2019 - 05/2026
Boston University	Boston, MA, USA
GPA 3.87 / 4.00	
B.Sc. in Computer Science, with distinction	09/2016 - 05/2019
American University of Beirut	Beirut, Lebanon
Major GPA 4.00 / 4.00	

PUBLICATIONS

	Vision-Language Framework for Multi-Sequence Brain Magnetic Resonance Imaging. MedRxiv.	2026
	Lteif, D. , Jia, S., Bit, S., Kaliaeve, A., Mian, A. Z., Small, J. E., ... & Kolachalama, V. B.	
	Higher-Order Domain Generalization in Magnetic Resonance-Based Assessment of Alzheimer's Disease. arXiv preprint arXiv:2601.01485 (under review).	2026
	Batool, Z., Lteif, D. , Kolachalama, V. B., Ozkan, H., & Aptoula, E.	
	Anatomy-Guided, Modality-Agnostic Segmentation of Neuroimaging Abnormalities. Human Brain Mapping 46(14), e70329.	2025
	Lteif, D. , Appapogu, D., Bargal, S. A., Plummer, B. A., & Kolachalama, V. B.	
	AI-based differential diagnosis of dementia etiologies on multimodal data. Nature Medicine, 1-13.	2024
	Xue, C.*, Kowshik, S.*, Lteif, D. , Puducheri, S., Zhou, O., Walia, A., ... & Kolachalama, V. B.	
	Disease-driven domain generalization for neuroimaging-based assessment of Alzheimer's disease. Human Brain Mapping, 45(8), e26707.	2024
	Lteif, D. , Sreerama, S., Bargal, S. A., Plummer, B. A., Au, R., & Kolachalama, V. B.	
	VISDA 2022 challenge: Domain adaptation for industrial waste sorting. In NeurIPS 2022 Competition Track (pp. 104-118). PMLR	2023
	Bashkurova, D., Mishra, S., Lteif, D. , Teterwak, P., Kim, D., Alladkani, F., Akl, J., Calli, B., Bargal, S.A., Saenko, K., Kim, D., Seo, M., Jeon, Y., Choi, D., Ettegui, S., Giryes, R., Abu-Hussein, S., Xie, B. Li, S.	
	Ani-GIFs: A benchmark dataset for domain generalization of action recognition from GIFs. Frontiers in Computer Science, 4. doi:10.3389/fcomp.2022.876846	2022
	Majumdar, S. S.*, Jain, S.*, Tourni, I. C.*, Mustafin, A., Lteif, D. , Sclaroff, S., Saenko, K., Bargal, S. A.	

SKILLS

- Machine Learning, Deep Learning, Statistical Modeling, Explainable AI, Multimodal AI, Computer Vision, Multimodal Large Language Models (MLLMs), Image Processing, Medical Imaging, Big Data, Data Processing, Data Visualization, Pattern Recognition, Feature Extraction, Performance Optimization, High-Performance Computing, Multi-GPU Training, Distributed Training, Algorithms, Data Structures
- Programming Languages
- Python, Java, C, C++, C#, JavaScript, PHP, Matlab, SQL
- Relevant Tools
- GitHub, Hugging Face, TensorFlow, PyTorch, Keras, Tensorboard, Weights & Biases, AWS Cloud

KEY ACHIEVEMENTS

-  Grace Hopper Celebration Conference Award
Boston University 2020
-  Dean's Honor List
American University of Beirut 2016-2019
-  First place winning team in the ACM Lebanese Collegiate Programming Contest
Beirut, Lebanon 2017
-  Fifth place winning team over Lebanon in the IEEEExtreme Programming Competition
2017

GRADUATE COURSEWORK

- CS585: Image and Video Computing
- CS542: Machine Learning
- CS640: Artificial Intelligence
- CS530: Advanced Algorithms
- CS535: Complexity Theory
- CS552: Operating Systems
- CS511: Formal Methods

EXPERIENCE

Graduate Research Fellow	01/2020 - Present
Boston University	Boston, Massachusetts, USA
- Designed and managed end-to-end (E2E) machine learning pipelines in Python and PyTorch for multimodal medical imaging and vision-language projects. - Built and evaluated deep learning models on large real-world clinical datasets, producing robust solutions for classification, segmentation, and report generation. - Performed large-scale data processing, statistical analysis, and visualization on big data from clinical and imaging sources. - Supported interdisciplinary teams by translating research models into practical workflows and providing technical guidance on model development and evaluation. - Demonstrated leadership across research projects involving robust representation learning, domain generalization, and multimodal AI. - Applied strong fundamentals in algorithms, coding, and data structures to develop scalable and efficient ML systems.	
Applied AI/ML Scientist Intern	06/2022 - 09/2022
Inari Medical Inc.	Irvine, CA, USA
- Developed and supported AI/ML solutions for intravascular ultrasound imaging used in minimally invasive endovascular procedures. - Enhanced imaging-analysis workflows by applying deep learning methods for clinically relevant feature extraction and decision support. - Managed data processing, model development, and evaluation tasks to produce practical solutions for real-world healthcare applications.	
Software Development Intern	08/2018 - 09/2018
PinPay SAL, Beirut Digital District	Beirut, Lebanon
Worked on Amazon Alexa, Google Assistant services and front-end web development.	
Graduate Teaching Fellow	09/2019 - Present
Boston University	Boston, Massachusetts, USA
CS511: Formal Methods I (Fall 2025) CS585: Image and Video Computing (Spring 2025) CS542: Machine Learning (Fall 2021) CS330: Introduction to Algorithms (Spring 2020) CS103: Introduction to Internet Technologies and Web Programming (Fall 2019)	
Instructor	07/2018
Future Developer Summer Camp, American University of Beirut	Beirut, Lebanon
Taught Unity and iOS development to students between the ages 12 and 18	

PROFESSIONAL ACTIVITIES

Reviewer	2024 - Present
IJCV, Frontiers in Neuroimaging, CVPR, CAI, Alzheimer's & Dementia Journal	
Seminar Co-chair	09/2021 - 06/2022
BU Image and Video Computing Group	Boston University
Invite speakers and organize weekly seminars at the IVC group at Boston University.	
Guest Speaker	08/2021
BU AI4ALL Program	Boston University
Gave a talk to high school students about Deep Learning and Explainable AI.	
Graduate Student Advisor	2020 - 2021
BU ACM-W Student Chapter	Boston University
Coordinated the student mentorship program and advocated for full engagement of women in computing and technology.	
Competitive Programming Contestant	11/2017
ACM Arab Collegiate Programming Competition 2017	Egypt
Teamed with fellow AUB students to compete in the 9th Arab Collegiate Programming Competition and ended up in the 31st place in the MENA region out of 104 participating teams.	

SELECTED COURSE PROJECTS

CS585 Final Project: Image Deblurring Applied to Recycling Data	
2021	Boston University
Applied and evaluated existing classical and deep learning deblurring methods to a video of recycling trash moving on a conveyor belt.	
CS542 Final Project: Image Classification on Covid-19 X-rays	
2020	Boston University
Implemented binary and multi-class classification using VGG and ResNet backbone architectures in Keras on a Covid-19 X-ray dataset.	
CS640 Final Project: Facial Expression Analysis Based on Videos of Presidential Candidates	
2019	Boston University
Worked with classmates to extend existing state-of-the-art deep convolutional neural networks for sentiment analysis on video clips of the 2020 Democratic Presidential candidates.	
Undergraduate Final Year Project: Emotion Recognition in an Uncontrolled Environment	
2019	American University of Beirut
Created an Android Application that predicted users' emotional state based on collected readings from ShimmerV3 Biophysical sensors and user self-assessment (in collaboration with the Psychiatry Department at AUBMC).	